

Enhancing Virtual Material Perception with Vibrotactile and Visual Cues

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Abstract

The ability to feel is crucial for a more realistic Virtual Reality (VR) experience. This research demo presents a way to enhance the VR experience through the use of a lightweight glove that provides vibrotactile feedback, allowing the user to feel the stiffness of virtual objects of different materials. The glove uses dielectric elastomer sensors and piezoelectric vibrotactile actuators, and is able to accurately convey the sensation of compressing various real-world objects through vibrotactile and visual cues. This demonstration presents a novel combination of visual information and vibrotactile feedback that has been shown to significantly improve a user's engagement with a VR environment

INTRODUCTION

Our hands play a vital role in our daily activities, so it is important to accurately replicate them in Virtual Reality (VR) to enhance immersion and improve interaction with virtual objects. Haptic feedback is a key factor in enhancing engagement when using VR. Previous research has demonstrated a variety of different ways to present haptic feedback using gloves, such as exoskeleton[1] and vibration[2].

In this demonstration we show a novel system for glove based haptic feedback. We have developed an ultra-lightweight vibrotactile system using a motion capturing glove and piezoelectric haptic actuators (Figure 1). Compared to other previous glove-based haptic system, our research has a number of novel elements, including:

- Using hardware that supports a wide range of vibrotactile feedback
- A unique method for transcribing physical stiffness into vibrotactile cues
- A combination of graphic and haptic feedback to create a visuo-haptic illusion that increase the haptic sensation.

In the rest of this paper we first describe in more detail the underlying technology that we develop, and then the user experience developed to demonstrate the technology.

HARDWARE AND SOFTWARE TECHNOLOGY

The prototype we have developed has three key components: 1) Glove hardware, 2) Algorithms for mapping physical stiffness into vibration cues, and 3) Graphics cues to create a visuo-haptic illusion.

The glove uses soft dielectric elastomer sensors to capture the movement of the fingers[3], while the location of the hand is tracked by the HTC Vive lighthouse and tracker system. The piezoelectric haptic actuators are able to produce a wide range of frequencies (20Hz to 400Hz), allowing them to accurately recreate various vibrotactile sensations[4]. A custom haptic controller board was designed to control the amplitude and frequency of the feedback.

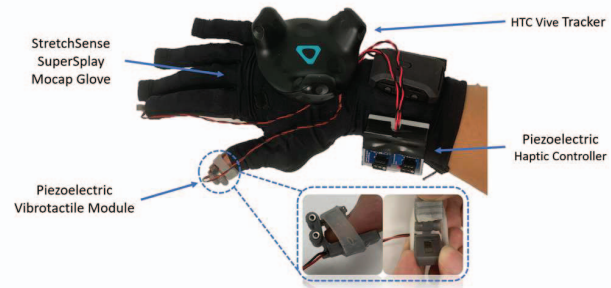


Figure 1 An overview of the lightweight vibrotactile glove

We developed a unique method for translating physical stiffness into vibrotactile cues. To do this, we tested various real-world materials (an aluminium block, a tennis ball, a silicone elastomer, and a balloon) using a load-testing machine (Figure 2.2) and used a polynomial regression model to recreate the haptic intensity of each material. The measured compression force was normalized so they can be used to control the intensity of the piezoelectric actuators (Figure 2.4). Additionally, each material was represented by a different vibration frequency, with the softest balloon being rendered at 20Hz and the hard aluminium block being rendered at 180Hz[5].

Finally, our approach relies on a visuo-haptic illusion[6], [7], where the appearance of the virtual objects changes based on how much they are squeezed (Figure 3). For example, a balloon will visibly deform when squeezed, while a metal ball will not. This helps to provide a more realistic representation of the object's softness[8].

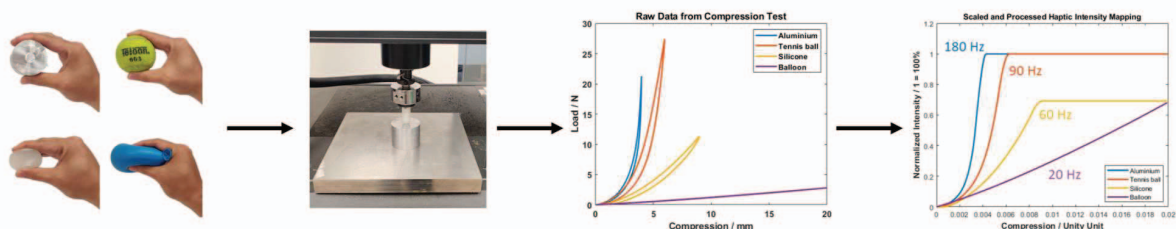


Figure 2 The process to transcribe real-world object's material properties to haptic feedback signals

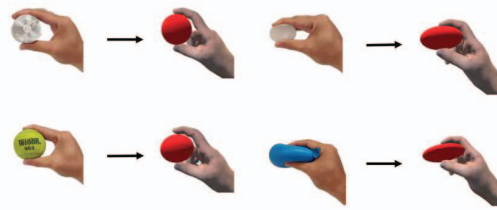


Figure 3 Visuo-haptic illusion of the four different materials

TECHNOLOGY DEMONSTRATION

The technology will be demonstrated using a simple VR interface where people can touch virtual balls, under conditions (Figure 4); 1) no visual feedback, haptic feedback only (H), 2) Haptic feedback with fingers penetrating the object (HPV), 3) Haptic feedback with object deformation (HDV).

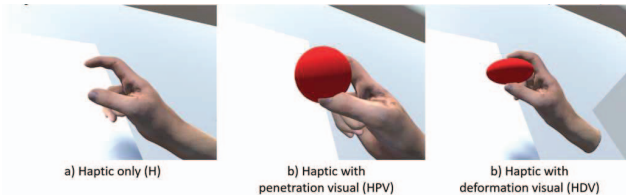


Figure 4 The three conditions of the demo

Users will put on a VR head mounted display and wear the prototype glove (one right hand glove). In the VR environment they will see virtual hands that mimic the motion of their real hands. On a virtual table in front of them they will see a dozen virtual balls and a box that they can place the balls into. They can try a task where they use their virtual hand to pick up and place the balls in the box as fast as they can. The demonstrator will switch between the three interface alternatives (H, HPV, HDV) so that they can see which condition provides the most realistic haptic feedback and best performance.

Our user studies indicate that the participants performed better with more visual information added to the haptic interaction; however visual information does not influence the perceived stiffness of objects. In a previous user study [Poster submission 1164] we found a that the HDV condition scored significantly higher in terms of user performance, and provided least effort.

In summary, the demonstration shows the ability to accurately recreate the sensation of compressing different materials in VR. By using our haptic transcription methods and visuo-haptic illusions, we are able to significantly enhance the user's engagement with virtual objects.

CONCLUSIONS AND FUTURE WORK

This paper describes the demonstration of a prototype haptic glove with visual feedback, that uses a visuo-haptic illusion to create more realistic haptic feedback in VR. This could be applied in many areas, such as virtual training, teleoperation and remote collaboration.

However, further research needs to be conducted to compare the user experience with only visual deformation to our current results, and to investigate the impact of mismatched visuo-haptic illusions (e.g. an object that looks like a balloon but feels like metal).

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